James Espinosa

[redacted] | GitHub | Portfolio

Education

University of Nottingham, BSc Computer Science (Hons) **Sep 2025 — Jul 2028**

- 75% Average in Semester 1, On track for first class

Experience

Aspiring Professionals in STEM (APSTEM) **Jan 2026 - Present**
Technology Ambassador

- Tech ambassador to a non-profit organisation, helping **1000+** students from underrepresented backgrounds/communities
- Hosted online taster workshops on First Year Computer Science at University

University of Nottingham **Jan 2026 - Present**
Coding Club Mentor

- Mentoring Students on Competitive Programing questions, going over "Easy Leetcode Questions"

ARM **Mar 2024**
Work Experience

- Coded a microbit for a challenge where we included responsive inputs for interaction

Projects

GRID (HackLondon '26 - 1st Place) | Python, CesiumJS

- Worked as a Team of 4 to display a satellite 3D map of earth, displaying electricity usage in areas in the East Midlands as circles from green,amber,red.
- Fullstack Engineer, integrating API's from Gemini, National Grid, Cesium
- Generates a PDF report by sending data to Gemini Flash 2.5 and displays a general overview
- Created plottable data centres & wind farms to predict possible power outages & sustainability

Snowball Stackers (UoN GameJam '25, 1st Place) | Unity, C#

- Worked as a team of 4, in charge of design & physics, with conditions & rolling
- Did extensive testing where I let my teammates create and polished different levels, then tested it to see how high the snowballs should stack to & how sensitive they should be per level

Gatwick GO! (HackSussex '26) | JavaScript

- Led a team of 4 to create Augmented Reality to scan moving planes in Gatwick & rewards users with upgrading economy flights to business flights
- Full Stack Engineer

Property Trend Forecaster (UCL RealTech Hack '26) | Python, streamlit, pandas, numpy

- Led a team of 4 to create a property trend forecaster via machine learning
- Full Stack Engineer, integrating ScanSans API to get UK area names and postcodes, allowing users to search it up which displayed current prices, while training an AI model, Catboost, to display future cheap/exorbitant prices

Paladins of Pi (HackNotts '25) | Python, json, streamlit, ollama

- Worked as a Team of 4 to create a working AI Chatbot running through Raspberry Pi
- Lead design on front-end, creating a UI to input a text prompt, linking it to back-end and Ollama, displaying a response with a .json file to display all chat history from a medieval Chatbot

Query a Degree | Python, tkinter, sqlite3, hashlib, csv

- Predicts the users probability of getting into any university or Degree Apprenticeship, using my school's and made up data (due to GDPR rules) saved onto a CSV file and successful grades
- Personal information was hashed using **SHA256**, avoiding plaintext for hackers to easily see

Personal Portfolio | HTML, CSS, JavaScript

- Holds all of my projects, and made to learn outside of university

Skills, Activities, Hobbies

Languages: English (Native)
Competitions: **204th** in the **UKIEPC**, **21st/34** teams in UoN's Coding Competition, **1st** place in **HackLondon '26**
Programming: Java, Python, SQL, HTML, CSS, JavaScript, C
Developer Tools: Git, GitLab, GitHub, Visual Studio Code, Jupyter, ILSpy
Interests: Reading fictional books, cooking.